

Motivation:

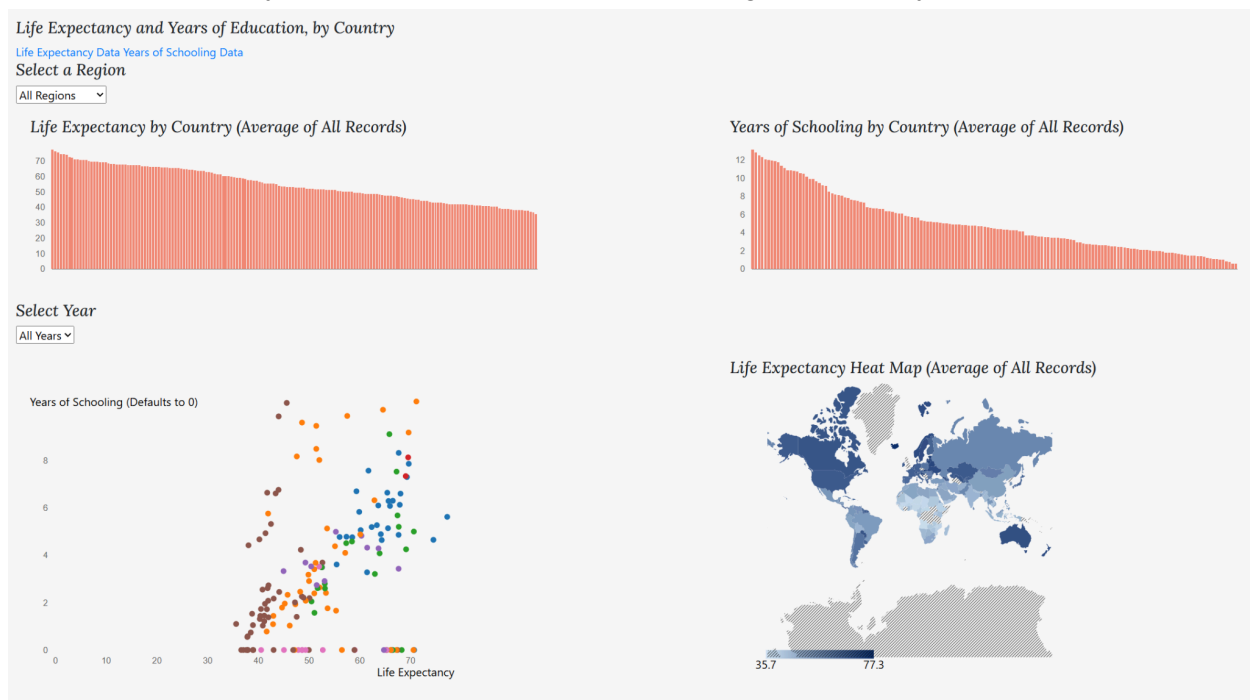
I think that it's really interesting to see how education correlates to life expectancy, and I think being able to see the data is cool. Honestly I just thought the two datasets, years of education and life expectancy, to be interesting to see.

Data:

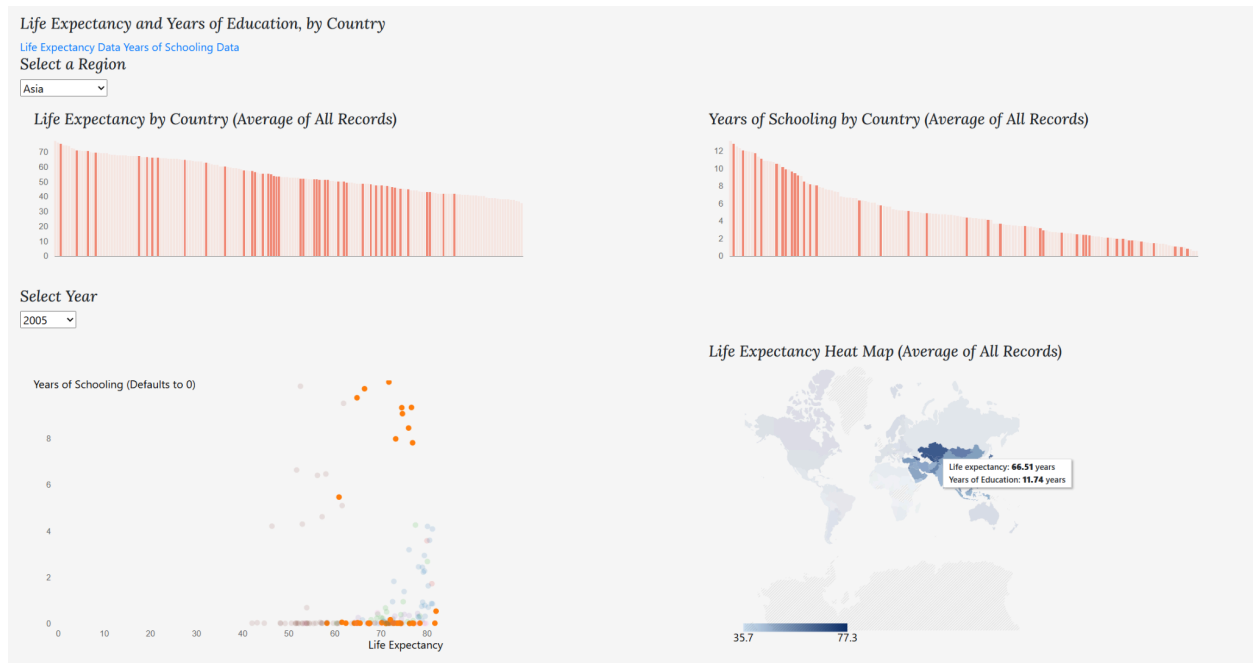
I used the [Life expectancy](#) data and [Average years of schooling](#) from Our World in Data. Both divide the data by country, and each data point also has the region it's associated with. Both have multiple years you can scroll through. Naturally, the values for both averages drop as you go back in time.

Views/GUI:

All of the views/charts fit on the screen at the same time. The top two charts are barcharts showing the average Life Expectancy and Years of Schooling, with the average being across all records. The bottom two are the scatterplot and the heatmap. The scatterplot allows you to select the year you want to view, "All Years" is the average. The dots are also colored along the region they belong to, and since there were some differences in the recorded values per country between the datasets, no data for the years of education default to zero. The heatmap scales off of the life expectancy, this one is also limited to the average across all years.



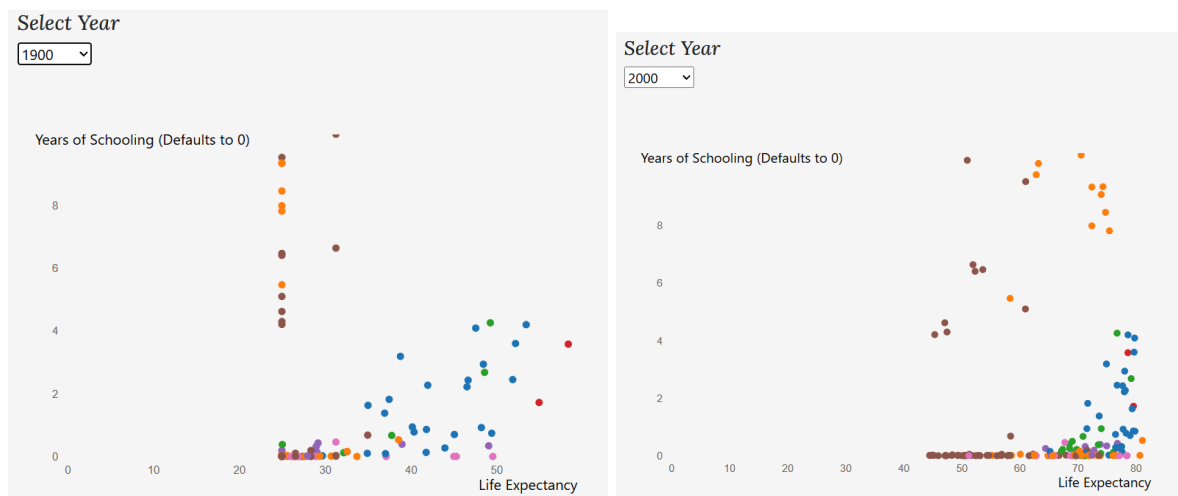
The interactivity (shown in the image below) shows if you select the region it will only show those records across all of the charts. The tooltip shown in the heat map is present on all of the charts, I'm only showing the heat map one for simplicity. The other records will also show the country name when you mouseover it.



Discovery:

Discovery was much easier with the heat map. It really just drove home how most “First World” countries have higher life expectancies on average. Even more when you look into more recent years with the scatterplot. However, I thought years of schooling would increase with the increase in life expectancy, overall. However, when you look at the scatterplot and click through it the life expectancy increases, but the years of education remain rather stagnant.

Ex:



Process:

I definitely took advantage of the examples provided by the professor (d3 examples specifically). So I utilized that primarily to create each of my charts, then fiddled around with them in order to get the desired result. So a lot of the library usage was decided for me, but the application uses bootstrap, D3, and jQuery. [Code](#) and [live application](#).

Challenges:

I definitely started doing the project too late in the timeline. If I had started even a week before I would have had the time to add on a lot of nicer quality of life features, such as a slide, cleaner charts, better backgrounds, etc. I feel like I learned a decent amount about how data is taken, filtered, and displayed. In addition to how these charts display different data better than others.

(Ex: Scatterplot was much better showing both datasets than the others).

In the future I think I'll make a much more concerted effort to begin work earlier than usual. Even if I'm not doing a whole lot at one time, it leaves time for me to think on what I want to do and how to do it. It would also give me the chance to better absorb how these data visualizations work under the hood.

AI Usage:

I tried to use this project as a way to experiment with AI usage by relying on it heavier than I usually do. I did this by using AI to help me rework the examples provided by the professor to fit with my data sets and how I wanted to explain them. It was also my first time using Copilot (It was built into VS Code so I figured I'd try it.) It definitely wasn't as helpful as I expected. It lost context pretty quickly and I would need to ask it to reread files where it was suggesting code changes that were already present. I also ran out of free tokens for the month within a day of using it. That was just for part 1. The later parts I only really used Claude.ai for error messages I couldn't make sense of. However, it did suggest some nice console.log commands that I hadn't thought of, and overall guided me on using the console log in the webpage that I really enjoyed. After I was pretty much finished with the code changes, I used Claude to quiz me on the code to help ensure I understood it. It also helped clear up some misunderstanding when I answered it wrong. I'll definitely be following that whenever I use Claude from now on, I feel like it helped a lot with my understanding.